

Name _____ Period _____

Skill 37.1 Exercise 1

For each scenario, decide whether it sounds more like *traditional programming* or *machine learning*.

Scenario	Traditional Programming or Machine Learning?	Explain
A program calculates the average of five test scores using a formula.		
A program learns from thousands of emails marked spam or not spam, then predicts whether a new email is spam.		
A program prints “eligible” if a person is at least 18 years old.		
A music app studies listening history and recommends songs a user might like.		

Skill 37.2 Exercise 1

A streaming app has the following dataset.

minutes_watched	number_of_likes	previous_genre	liked_new_show
120	8	comedy	yes
45	1	drama	no
200	12	comedy	yes
80	3	action	no

What is the label?

What are three possible features?

What does each row represent?

Why would the app want to train a model using this data?

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Skill 37.3 Exercise 1

For each dataset, decide whether it is **labeled** or **unlabeled**.

Dataset Description	Labeled or Unlabeled?	Explain
Emails with sender, number of links, subject length, and spam/not spam.		
Customer shopping data with purchase totals and visit frequency, but no customer group assigned.		
Houses with square footage, number of bedrooms, and sale price.		
Songs with tempo, energy, danceability, and no genre label.		

Skill 37.4 Exercise 1

For each situation, identify whether it is **supervised learning** or **unsupervised learning**.

Scenario	Supervised or Unsupervised?	Explain
A model predicts whether a patient has a disease using past patient records that include disease status.		
A company groups customers into unknown categories based on shopping behavior.		
A model predicts the sale price of a used car using data from cars that already sold.		
A social media app finds groups of users with similar viewing habits.		

Skill 37.5 Exercise 1

For each task, identify whether it is **regression**, **classification**, or **clustering**.

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Task	Regression, Classification, or Clustering?	Explain
Predicting the number of points a basketball player will score.		
Predicting whether a customer will cancel a subscription: yes or no.		
Grouping students based on study habits without knowing the groups ahead of time.		
Predicting the price of a used phone.		
Predicting whether a review is positive, negative, or neutral.		

Skill 37.6 Exercise 1

For each scenario, identify:

- whether the data is labeled or unlabeled
- whether the problem is supervised or unsupervised
- whether the task is regression, classification, or clustering

Scenario	Labeled or Unlabeled?	Supervised or Unsupervised?	ML Task
A real estate website predicts house prices using past home sales.			
A school predicts whether students are likely to pass a course using past student records labeled pass/fail.			
A company groups customers based on shopping habits without any pre-existing group names.			
A weather app predicts tomorrow's high temperature.			
A model predicts whether a text message is spam or not spam.			